

Project Title : 48

Suraksha-Setu (Infrastructure)

Android App Development using GenAI

Comprehensive Project Requirement Document

1. Project Overview

Project Title	Suraksha-Setu – A Hyper-Local Safety Network Android App
Project Number	48
Program	MindMatrix VTU Internship Program
Domain	Public Safety / Women's Safety / Rural Infrastructure / GenAI
Tech Stack	Android Studio Kotlin Jetpack Compose Firebase (FCM) Google Geofencing API SensorManager GenAI
Objective	Suraksha-Setu is a Hyper-Local Safety Network app that connects users to a 'Circle of Trust' (neighbors, shopkeepers, and volunteers) within a 500m radius. When triggered, it sets off a loud alarm on the phones of nearby Verified Volunteers for immediate physical help — designed specifically for women and elderly citizens in rural areas.

Core Idea: "Empower rural communities to protect their own — connecting users to trusted local volunteers within seconds."

2. The Problem Statement

In rural areas, women and elderly citizens walking alone at night often feel unsafe. Traditional SOS apps notify the police, but in a village, the 'First Responders' are usually neighbors or local youth groups. Police response times in rural areas can be dangerously slow, leaving vulnerable people without immediate physical help when they need it most.

Key Pain Points

- Women and elderly face safety threats while walking alone, especially at night.
- Traditional SOS apps alert only police — response time in rural areas is too slow.
- No existing app leverages hyper-local community networks for instant physical help.
- Lack of a verified volunteer system to ensure trustworthy first responders.
- Limited tech-based neighborhood watch culture in rural communities.

3. Detailed Description — The Vision

Suraksha-Setu is a 'Hyper-Local Safety Network.' It connects the user to a 'Circle of Trust' (neighbors, shopkeepers, and volunteers) within a 500m radius. When triggered, it sets off a loud alarm on the phones of nearby 'Verified Volunteers' for immediate physical help.

Unlike traditional SOS apps that rely solely on emergency services, Suraksha-Setu builds a real-time community response layer — turning ordinary citizens into first responders. By combining geofencing, push notifications, shake gestures, and live audio streaming, it creates a safety net that is faster, more

personal, and more relevant to rural contexts.

4. App Usage & User Flow

Core Features

Safe-Circle: Add up to 5 trusted neighbors or family members to your personal safety circle. These contacts receive priority alerts when an SOS is triggered.

Panic Trigger: Shake the phone or double-tap the power button to instantly alert the Circle of Trust. Works even when the screen is locked or the app is in the background.

Live Audio: Automatically records and streams 30 seconds of audio to Circle members upon SOS trigger — providing real-time situational awareness.

Volunteer Mode: Local youth and community members can register as 'Verified Responders.' They receive geofenced alerts whenever a nearby SOS is triggered.

Step-by-Step User Flow

- 1. User installs app and registers with phone number and name.
- 2. User sets up Safe-Circle by adding 5 trusted contacts.
- 3. User optionally enables Volunteer Mode to become a Verified Responder.
- 4. In an emergency: user shakes phone or double-taps power button.
- 5. App instantly sends FCM alerts to all Safe-Circle members.
- 6. Google Geofencing API identifies all Verified Volunteers within 500m and triggers loud alarm on their devices.
- 7. Live audio (30 seconds) is streamed to Safe-Circle members.
- 8. GPS coordinates of the user are broadcast to all responders.
- 9. Volunteers navigate to user location using embedded map view.
- 10. Incident is logged with timestamp, location, and responder details.

5. Functional Requirements

5.1. User Registration & Profile

- Phone number-based registration with OTP verification.
- Profile setup: name, address, emergency contact details.
- Option to register as a 'Verified Volunteer / Responder'.
- Volunteer verification badge after community approval.

5.2. Safe-Circle Management

- Add up to 5 trusted contacts to personal Safe-Circle.
- Contacts receive invite notification to join the app.
- View and manage Safe-Circle members from the dashboard.
- Each Safe-Circle member's availability status is visible.

5.3. SOS / Panic Trigger

- Shake gesture detection using SensorManager (works in background).
- Double-tap power button shortcut for SOS activation.
- Large, high-visibility SOS Button on the home screen.
- Confirmation screen with 3-second cancel option to prevent false alarms.
- SOS triggers simultaneously: FCM alerts + geofenced volunteer alarms + live audio stream.

5.4. Live GPS & Location Sharing

- User's live GPS coordinates broadcast to Safe-Circle and nearby volunteers.
- Location updates every 10 seconds during active SOS.
- Map view showing user pin and responder locations.
- Alert includes Google Maps deep link for one-tap navigation.

5.5. Live Audio Streaming

- Auto-record 30 seconds of ambient audio on SOS trigger.
- Stream audio in real-time to Safe-Circle members via Firebase.
- Audio stored temporarily (auto-deleted after 24 hours) for privacy.
- Responders can listen to live audio from the alert notification.

5.6. Volunteer / Verified Responder Mode

- Any community member can register as a Verified Responder.
- Geofencing API monitors for SOS events within 500m radius.
- Responder's phone triggers loud alarm when a nearby SOS is detected.
- Responder can accept or decline the alert — accepted count visible to user.
- Responder location shared with SOS user after accepting.

5.7. Incident Log

- All SOS events logged with timestamp, location, and responder details.
- User can view personal incident history.
- Incident report shareable with authorities if required.

5.8. GenAI Safety Advisor

- Gemini API integration for intelligent safety tips based on time and location.
- AI generates personalized safety route suggestions for frequent paths.
- Post-incident AI summary: 'Here is what happened and how to stay safer next time.'
- Appears as an AI Advisor card on the home dashboard.

6. Technical Implementation & Hints

Component	Technology	Purpose
Push Alerts	Firebase Cloud Messaging (FCM)	Instant, high-priority SOS notifications to Safe-Circle and volunteers
Geofencing	Google Geofencing API	Alert only Verified Volunteers within a 500m radius of SOS trigger
Shake Detection	Android SensorManager	Detect emergency shake gestures even in background/locked state
Live Location	Google Fused Location API	Real-time GPS coordinates broadcast during SOS event
Audio Stream	Android MediaRecorder + Firebase Storage	Record and stream 30 seconds of live audio to contacts
UI Framework	Jetpack Compose	Modern declarative Android UI with high-visibility design
Database	Room DB + Firebase Firestore	Local persistence + cloud sync for user profiles and incidents
GenAI Layer	Gemini API (REST)	Intelligent safety tips, route suggestions, and post-incident summaries
Authentication	Firebase Auth (OTP)	Secure phone number-based user registration

Key Technical Challenges & Solutions

- Background SOS Trigger: Use Android Foreground Service + SensorManager to detect shake gestures even when app is in background or screen is locked.
- Geofenced Volunteer Alerts: Register geofences for all active volunteers using Google Geofencing API. SOS trigger queries volunteers within 500m and sends targeted FCM.
- Low-Latency Audio Streaming: Use Firebase Realtime Database for signaling + Firebase Storage for audio chunks to minimize streaming delay.
- Battery Optimization: Implement intelligent geofence batching and use WorkManager for background tasks to minimize battery drain on volunteer devices.
- False Alarm Prevention: 3-second countdown UI with cancel option and shake sensitivity threshold tuning to avoid accidental triggers.

7. Non-Functional Requirements

Requirement	Description
Performance	SOS alert must reach all contacts within 3 seconds of trigger.
Reliability	App must function during SOS even with poor network — local alarm still sounds.
Usability	High-visibility UI with large SOS button; usable by elderly and low-literacy users.
Security	Audio data auto-deleted after 24 hours; GPS sharing stops when SOS is cancelled.
Compatibility	Android 9.0 (API 28) and above.
Offline Support	Basic SOS alert to Safe-Circle via SMS fallback if internet is unavailable.
Scalability	Architecture supports city-wide volunteer network expansion.
Privacy	No sensitive data stored externally without explicit user consent.

8. System Architecture

Architecture Pattern: MVVM (Model-View-ViewModel) + Clean Architecture

The app follows MVVM with a clean separation of concerns across UI, business logic, and data layers — ensuring testability, maintainability, and scalability.

Layer	Technology	Role
View (UI)	Jetpack Compose	Renders screens; observes ViewModel state; high-visibility SOS UI
ViewModel	Kotlin ViewModel + StateFlow	Holds UI state; orchestrates SOS trigger logic
Repository	Repository Pattern	Single source of truth; mediates between DB, Firebase, and APIs
Local DB	Room Database	Stores user profile, Safe-Circle, and incident logs offline
Cloud Layer	Firebase FCM + Firestore + Storage	Real-time alerts, volunteer sync, and audio storage
Location Layer	Google Fused Location + Geofencing API	Live GPS tracking and volunteer proximity detection
Sensor Layer	Android SensorManager	Shake gesture detection for background SOS trigger
GenAI Layer	Gemini API (REST)	Safety tips, route advice, post-incident AI summaries

Data Flow

- SOS Trigger (Shake/Button) → SensorManager → ViewModel → Repository → FCM Service → Safe-Circle Devices
- Geofencing Event → Google Geofencing API → Firebase → Volunteer Devices (loud alarm)
- GPS Location → Fused Location API → Firebase Firestore → Responder Map View
- Live Audio → MediaRecorder → Firebase Storage → Safe-Circle Audio Player
- Sales/Incident Data → Repository → Gemini API → GenAI Advisor Card on Dashboard

9. Database Schema

Table	Fields	Description
User	id (PK), name, phone, is_volunteer, verified_status, fcm_token	Registered user and volunteer profiles
SafeCircle	id (PK), user_id (FK), contact_phone, contact_name, relationship	User's trusted contact circle
SOSEvent	id (PK), user_id (FK), triggered_at, location_lat, location_lng, status, audio_url	Log of all SOS incidents
Responder	id (PK), sos_id (FK), volunteer_id (FK), response_status, responded_at	Volunteer response records per SOS event
AIAdvice	id (PK), generated_at, advice_text, context_note	Cached Gemini API safety recommendations

10. UI/UX Guidelines

Design Goals

- Emergency-first design: The SOS button must be the largest, most prominent element on screen.
- One-tap activation: SOS must be triggerable within a single tap or shake — no navigation required.
- High-visibility: High-contrast colors (red/white) for immediate recognition under stress.
- Accessibility-first: Large touch targets and minimal text for elderly and low-literacy users.
- Trust-building UI: Show Safe-Circle member status (online/offline) on the dashboard.

Color Palette

Role	Color	Hex Code	Usage
Primary (Alert)	Emergency Red	#B71C1C	SOS button, alert states, headings
Accent	Deep Orange	#E65100	Highlights, active states, FAB
Safe State	Trust Green	#2E7D32	Safe-Circle active, volunteer confirmed
Warning	Amber	#FF8F00	Pending alerts, unconfirmed responders
Background	Light Grey	#F5F5F5	Screen backgrounds for readability
Text Primary	Near Black	#1A1A1A	All primary text content

Key Screens

- 1. Splash / Onboarding Screen – App logo, tagline 'Your Community. Your Shield.' + OTP login
- 2. Home Dashboard – Large SOS Button (center) + Safe-Circle status + AI Safety Tip card
- 3. SOS Active Screen – Countdown timer + Cancel button + 'Alerting your Circle...' status
- 4. Safe-Circle Manager – Add/remove trusted contacts + online status indicators
- 5. Volunteer Map Screen – Live map showing user pin + responding volunteer locations
- 6. Volunteer Mode Screen – Toggle to become Verified Responder + incoming alert view

- 7. Incident Log Screen – History of past SOS events with timestamps and responder details
- 8. Settings Screen – Profile management + sensitivity settings + About + Privacy policy

11. GenAI Integration — Gemini API

Suraksha-Setu integrates the Gemini API to provide intelligent, context-aware safety advice that goes beyond simple alerts — making the app a proactive safety companion, not just a reactive SOS tool.

GenAI Features

- Safety Tips Advisor: Generates real-time safety tips based on user's time of day and location type (e.g., 'It's late evening — ensure your Safe-Circle is updated before heading out').
- Post-Incident Summary: After an SOS event, Gemini generates a human-readable summary: 'Your alert was received by 3 volunteers. Responder arrived in 4 minutes. Here are safety tips for your route.'
- Route Safety Suggestions: Analyzes frequent travel patterns and suggests safer alternatives for night travel.
- Volunteer Briefing: When a volunteer accepts an SOS, Gemini generates a quick briefing with context about the situation to help them respond effectively.

12. Impact Goals

Women's Safety

Creating a secure environment for women in rural areas by connecting them to a verified community response network — reducing dependency on slow emergency services.

Social Cohesion

Strengthening the 'Neighborhood Watch' culture through technology — encouraging local youth to become active community protectors.

Emergency Efficiency

Reducing emergency response time by involving local citizens as first responders — dramatically cutting the gap between distress signal and physical help.

Digital Inclusion

Bringing AI-powered safety tools to rural communities with low-literacy-friendly design — making GenAI accessible to those who need it most.

Community Empowerment

Transforming passive bystanders into active Verified Volunteers — building a culture of collective responsibility for public safety.

13. Success Criteria for Students

- ✓ The 'Shake' gesture must trigger the alert even if the app is in the background.
- ✓ The alert must include the user's live GPS coordinates shared to Safe-Circle and volunteers.
- ✓ The UI must be high-visibility with a prominent, large SOS Button on the home screen.
- ✓ FCM push notification must reach Safe-Circle contacts within 3 seconds of SOS trigger.
- ✓ Geofencing must correctly identify and alert only volunteers within the 500m radius.
- ✓ Live audio must auto-record and stream for 30 seconds upon SOS activation.
- ✓ The GenAI Advisor card must display a contextually meaningful safety tip.
- ✓ The app must continue to function (local alarm + SMS fallback) without internet.

14. Deliverables for Students

- 1. Wireframe (Figma/Canva) – All 8 key screens with navigation flow and SOS interaction design
- 2. Database Schema Diagram – ER model showing all 5 tables and Firebase collections
- 3. Functional App (APK) – Working prototype with SOS trigger + geofencing + FCM alerts
- 4. Live Location Feature – GPS broadcast to Safe-Circle and volunteer map view
- 5. GenAI Integration – Gemini API safety advisor card on home dashboard
- 6. Project Report – Feature description, architecture diagram, screenshots, team roles, and code overview
- 7. Demo Video (2–3 min) – Screen recording demonstrating SOS trigger, volunteer alert, and AI advisor

15. Future Scope & Enhancement Ideas

- City-Wide Volunteer Network – Scale beyond 500m to district-level volunteer mapping
- Police Integration API – Auto-notify nearest police station with GPS on SOS trigger
- Wearable Integration – Trigger SOS from smartwatch or fitness band
- Offline Mesh Networking – Use Bluetooth mesh to relay SOS without internet
- Regional Language Support – Kannada, Hindi, Tamil for wider rural accessibility
- AI Threat Detection – Analyze audio for distress sounds and auto-trigger SOS
- SHG / Group Safety Mode – Shared safety network for women's self-help groups
- Community Safety Score – Dashboard showing volunteer density and response times per area

16. Evaluation Parameters

Category	Weightage
Functionality (SOS Trigger, Geofencing, FCM Alerts, Live GPS)	30%
UI/UX Design (High-Visibility SOS UI, Accessibility, Ease of Use)	25%
Code Quality & Structure (MVVM, Clean Architecture, Firebase)	20%
Innovation / GenAI Integration (Gemini API Safety Advisor)	15%
Presentation & Documentation (Report + Demo Video)	10%

17. Next Action Items for Teams

- Set up Firebase project — enable FCM, Firestore, Auth (OTP), and Storage.
- Obtain Google Maps API key and enable Geofencing API in Google Cloud Console.
- Obtain Gemini API key from Google AI Studio for GenAI integration.
- Prepare wireframes for all 8 screens in Figma — get mentor approval before coding.
- Divide responsibilities: UI Developer | Firebase/Backend | Sensor/SOS Logic | GenAI Integrator | Tester.
- Create Android project in Android Studio with MVVM + Clean Architecture folder structure.
- Implement Foreground Service for background shake detection as first milestone.